

Conference on Algebraic Representation Theory 2026

CART 2026

December 4 (Fri.) – 6 (Sun.), 2026

Okinawa Institute of Science and Technology (OIST)

Seminar Room L4E48, Okinawa, Japan

Titles and Abstracts

Speakers are listed in alphabetical order.

Katsuyuki Bando (Keio University)

Title: TBA

Abstract: TBA

Dong Jun Choi (Seoul National University)

Title: TBA

Abstract: TBA

Jiepeng Fang (The University of Hong Kong)

Title: Thickening realization and positivity properties of canonical bases

Abstract: For any symmetric Cartan datum, Lusztig conjectured that the canonical basis $\dot{\mathbf{B}}$ of the modified quantum group $\dot{\mathbf{U}}$ has positivity properties with respect to the multiplication and comultiplication. We introduce an algebraic method called the thickening realization to relate $\dot{\mathbf{B}}$ to the canonical basis of the negative part $\tilde{\mathbf{U}}^-$ of a larger quantum group. More precisely, it identifies the structure constants in $\dot{\mathbf{U}}$ with relevant structure constants in $\tilde{\mathbf{U}}^-$. We prove that $\dot{\mathbf{B}}$ has the positivity properties with respect to the comultiplication and bilinear form of $\dot{\mathbf{U}}$, as well as the multiplication whenever one factor is spherical parabolic. In particular, Lusztig's positivity conjecture holds for any simply-laced finite type. This is a joint work with Xuhua He.

You-Hung Hsu (National Cheng Kung University)

Title: TBA

Abstract: TBA

Woo-Seok Jung (University of Seoul)

Title: A Comparison of Cluster Algebra Structures Arising from i -Boxes and Demazure Weaves

Abstract: We compare two cluster algebra structures associated with a positive braid of finite ADE type. The first arises from a localized bosonic extension, whose initial seeds are constructed from admissible chains of i -boxes and are closely related to monoidal categorification. The second arises from the coordinate ring of a braid variety, with initial seeds constructed from Demazure weaves.

For each admissible chain of i -boxes, we explicitly construct a corresponding Demazure weave and show that the resulting seeds are identified under an algebra isomorphism between the localized bosonic

extension and the coordinate ring of the braid variety. Under this isomorphism, the PBW vectors are identified with the natural coordinates on the braid variety.

This is joint work with JiSun Huh, Myungho Kim, and Euiyong Park.

Chun-Ju Lai (Academia Sinica)

Title: Foundations and Applications of Quantum Wreath Products

Abstract: Quantum wreath products deform classical wreath product groups by allowing relation coefficients to take values in a tensor product algebra rather than the ground field. This framework unifies the study of structure and representation theory of algebras admitting a Bernstein-Lusztig presentation, including Vignéras' pro- p Hecke algebras for p -adic groups (and their mteaplectic analogs), Kleshchev-Muth's affine zigzag algebras, and Hu's "Hecke algebras" for the wreath products between symmetric groups. We will discuss recent progress of our theory in seemingly unrelated directions: the Ginzburg-Guay-Opdam-Rouquier conjecture on highest-weight covers, the realization of imaginary strata for affine KLR algebras in positive characteristic, the breakdown of the local Shimura correspondence at the pro- p level, and the categorification perspectives of towers of quantum wreath products. The talk is based on joint work with Buciumas (Pohang), Cailan Li (Academia Sinica), Minets (Bonn), Nakano (Georgia), and Ziqing Xiang (Shenzhen).

Donggun Lee (Korea Institute for Advanced Study)

Title: TBA

Abstract: TBA

Kang Lu (Southern University of Science and Technology)

Title: TBA

Abstract: TBA

Haruto Murata (The University of Tokyo)

Title: TBA

Abstract: TBA

Yung-Ning Peng (National Cheng Kung University)

Title: TBA

Abstract: TBA

Heizo Sakamoto (The University of Tokyo)

Title: TBA

Abstract: TBA

Pavel Turek (Okinawa Institute of Science and Technology)

Title: New decomposition columns in every block

Abstract: A central open problem in the modular representation theory of symmetric groups (and Hecke algebras) is the computation of decomposition numbers, which describe how characteristic-zero irreducible modules decompose in prime characteristic. These numbers are organised in a block matrix, called the decomposition matrix, whose structure remains a mystery. In this talk, we present new whole columns of this matrix labelled by a novel family of partitions called d-balanced partitions;

this seems to be the first description of whole columns in each block to date! Moreover, the column description is exceptionally elementary. The talk is based on results proved with David Hemmer (for $d = 2$) and with Bim Gustavsson, Stacey Law, Lorenzo Putignano and Liron Speyer (for $d > 2$).

Yumeng Wu (Peking University)

Title: TBA

Abstract: TBA